

Solutions to exercises in Lecture 9

Q1) The Lax-Friedrichs and leap frogs do not fit in the hierarchy based on Eq. (4) in the lecture notes.

This is beyond what we have covered in the lectures. However, there is a systematic way to determine the order of accuracy, for those of you interested.

For the Lax-Friedrichs, after Taylor expansion

$$\frac{u_i^{n+1} - \frac{1}{2}(u_{i+1}^n + u_{i-1}^n)}{\Delta t} = \left. \frac{\partial u}{\partial t} \right|_i + \frac{\Delta t}{2} \left. \frac{\partial^2 u}{\partial t^2} \right|_i - \frac{\Delta x^2}{2\Delta t} \left. \frac{\partial^2 u}{\partial x^2} \right|_i + \text{error terms at } \Delta t^2 \text{ and } \Delta x^2 \text{ order \& higher}$$

$$\frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} = \left. \frac{\partial u}{\partial x} \right|_i + \text{error terms at } \Delta x^2 \text{ order \& higher, that we can ignore}$$

substituting to

$$\frac{u_i^{n+1} - \frac{1}{2}(u_{i+1}^n + u_{i-1}^n)}{\Delta t} = \frac{u_{i+1}^n + u_{i-1}^n}{2} - c \Delta t \frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x}$$

$$\frac{u_i^{n+1} - \frac{1}{2}(u_{i+1}^n + u_{i-1}^n)}{\Delta t} + c \frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} = 0$$

$$\left. \frac{\partial u}{\partial t} \right|_i + \frac{\Delta t}{2} \left. \frac{\partial^2 u}{\partial t^2} \right|_i - \frac{\Delta x^2}{2\Delta t} \left. \frac{\partial^2 u}{\partial x^2} \right|_i + c \left. \frac{\partial u}{\partial x} \right|_i = 0$$

$$\left(\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} \right) \Big|_i + \left(\frac{\Delta t}{2} \frac{\partial^2 u}{\partial t^2} \right) \Big|_i - \left(\frac{\Delta x^2}{2\Delta t} \frac{\partial^2 u}{\partial x^2} \right) \Big|_i = 0$$

the hyperbolic PDE we want

leading error terms

this means the scheme is 1st order accurate in time

due to von Neumann stability analysis, $\Delta x \propto \Delta t$ for the scheme to be stable. Hence, $\frac{\Delta x^2}{\Delta t} \propto \Delta x$. scheme is 1st order accurate in space.

For the leap frog method,

$$u_i^{n+1} = u_i^{n-1} - c\Delta t \frac{u_{i+1}^n - u_{i-1}^n}{\Delta x}$$

$$\frac{u_i^{n+1} - u_i^{n-1}}{2\Delta t} + c \frac{u_{i+1}^n - u_{i-1}^n}{\Delta x} = 0$$

$$\left. \frac{\partial u}{\partial t} \right|_i^n + \frac{1}{2} (\Delta t)^2 \left. \frac{\partial^2 u}{\partial t^2} \right|_i^n + c \left. \frac{\partial u}{\partial x} \right|_i^n + c \frac{(\Delta x)^2}{2} \left. \frac{\partial^2 u}{\partial x^2} \right|_i^n = 0$$

$$\underbrace{\left(\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} \right)}_i^n + \underbrace{\frac{(\Delta t)^2}{2} \frac{\partial^2 u}{\partial t^2}}_i^n + \underbrace{\frac{c}{2} (\Delta x)^2 \frac{\partial^2 u}{\partial x^2}}_i^n = 0$$

hyperbolic PDE
of interest

leading error term
2nd order in time

leading error term
2nd order in space